PROJECT DESIGN PHASE
Proposed Solution

Date	01 JULY 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	2 Marks

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1. Introduction

The **Personalized Networking Assistant (Smart Connect AI)** is an AI-powered web application developed to assist students, professionals, researchers, entrepreneurs, and conference participants in preparing for networking events. The application analyzes event descriptions, generates intelligent conversation starters, verifies knowledge using the Wikipedia API, stores networking session history, and collects user feedback through a user-friendly web interface.

The solution integrates **Artificial Intelligence (AI)**, **Natural Language Processing (NLP)**, **FastAPI**, **Streamlit**, and the **Wikipedia API** to provide an intelligent networking assistant that enhances communication confidence and professional networking experiences.

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2. Proposed Solution Overview

The proposed system follows a **client-server architecture**.

- The **Streamlit frontend** provides an interactive interface where users can enter event descriptions, search topics, save networking sessions, and submit feedback.
- The **FastAPI backend** receives user requests and performs event analysis, conversation generation, knowledge retrieval, session management, and feedback processing.
- The **Wikipedia API** supplies verified information for networking topics.
- **JSON files** are used to store networking history and user feedback locally.

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3. System Architecture

The proposed architecture consists of five major components:

1. User Layer

Users interact with the application by providing:

- Name
- Profession
- Networking Goal

- Event Description
- Knowledge Topic
- Feedback

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2. Frontend Layer (Streamlit)

Responsibilities include:

- Event Description Input
- Generate Conversation Starters
- Knowledge Assistant
- Save Networking Session
- Feedback Form
- Display Results
- User Notifications

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3. Backend Layer (FastAPI)

The backend contains several services:

Event Analyzer Service

- Analyze event description
- Extract important networking topics

Conversation Generator Service

- Generate AI-powered conversation starters
- Return personalized suggestions

Knowledge Assistant Service

- Connect with Wikipedia API
- Retrieve verified summaries

History Logger Service

- Save networking sessions
- Store event information

Feedback Logger Service

- Save user feedback
- Maintain feedback records

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4. External Service

Wikipedia API

Provides:

- Topic verification
- Trusted information
- Short summaries
- Reliable knowledge retrieval

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5. Data Storage

The application stores data using JSON files.

history.json

Stores

- Event Description
- Extracted Topics
- Date & Time

feedback. json

Stores

- User Feedback
- Date & Time

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4. Workflow of the Proposed Solution

The overall workflow of the application is as follows:

Step 1	User opens the Smart Connect AI web application.
Step 2	User enters a networking event description.
Step 3	The Streamlit frontend sends the request to the FastAPI backend.
Step 4	The Event Analyzer extracts important networking topics.
Step 5	The Conversation Generator creates personalized conversation starters.
Step 6	Generated conversation starters are displayed to the user.
Step 7	The user searches any topic in the Knowledge Assistant.
Step 8	The backend requests verified information from the Wikipedia API.
Step 9	The verified summary is displayed to the user.
Step 10	The user saves the networking session.
Step 11	The application stores networking history in **history. json**.
Step 12	The user submits feedback.
Step 13	Feedback is stored in **feedback. json**.

5. Advantages of the Proposed Solution

The proposed system provides several advantages:

- AI-powered networking assistance
- Personalized conversation starters
- Reliable knowledge verification
- Interactive web interface
- Faster networking preparation
- Easy session management
- Local data storage
- User feedback collection
- Modular system architecture
- Scalable and maintainable design

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6. Technologies Used

Layer	Technology
Programming Language	Python
Frontend	Streamlit
Backend	FastAPI
AI Processing	Natural Language Processing (NLP)
Knowledge Source	Wikipedia API
API Communication	Requests
Data Storage	JSON
Server	Uvicorn
IDE	Visual Studio Code
Version Control	Git & GitHub

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7. Expected Output

The proposed solution enables users to:

- Generate intelligent conversation starters.

- Verify networking topics using Wikipedia.
- Save networking sessions.
- Submit feedback.
- Improve networking confidence.
- Prepare effectively for conferences, seminars, workshops, and career fairs.

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8. Future Enhancements

The system can be extended with:

- OpenAI GPT integration
- Voice-based conversation assistance
- LinkedIn profile integration
- User authentication
- Cloud database support (MongoDB/Firebase)
- Multi-language support
- AI recommendation engine
- Mobile application support

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9. Conclusion

The **Personalized Networking Assistant (Smart Connect AI)** provides a comprehensive AI-powered solution for networking preparation. By integrating **FastAPI**, **Streamlit**, **Natural Language Processing**, **Wikipedia API**, and **JSON storage**, the application offers intelligent conversation generation, trusted knowledge verification, networking history management, and user feedback collection. The proposed solution improves communication confidence, simplifies networking preparation, and creates a scalable platform that can be enhanced with advanced AI capabilities in the future.